MIL-T-0036347A (DSA-DM)
24 August 1973
USED IN LIEU OF
MIL-T-36347 (DM)
22 March 1965

MILITARY SPECIFICATION

TEST BRIDGE

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-T-36347 (DM). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of test bridge and conductivity cell suitable for the measurement of solution conductivity from 500 to 7000 micromhos per centimeter.

* 2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6630

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1125-64 Electrical Conductivity of Water, Tests for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and suitable in all respects for the purpose intended.

3.2 Design. The test bridge shall be a compact and rugged instrument suitable for operating on 115 volt, 60 cycle, alternating current. The measuring circuit shall be a Wheatstone bridge and shall have a vacuum tube null indicator. The bridge shall have a manual temperature compensator adjustable from 64° to 190° Fahrenheit (°F.).

3.2.1 Cell component. The bridge shall be supplied with a dip-type conductivity cell constructed of high-temperature polystyrene with platinized gold-plated nickel rod electrodes. The cell shall have a constant of 2.

3.3 Performance. The test bridge shall read from 500 to 7000 micromhos per centimeter specific conductance referred to 25° Centigrade (°C.), and shall have a midpoint of 1500 micromhos per centimeter at which optimum sensitivity is found, when tested as specified in 4.4.1.

* 3.4 Accuracy. Accuracy of the conductivity reading shall be ± 4%, when tested as specified in 4.4.1.

* 3.5 Case. The bridge shall be housed in a sheet metal case finished with enamel.

3.6 Identification marking. Each test bridge shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

* 3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete

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instructions for installation, operation and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

* 3.8 Workmanship. Workmanship shall be first class throughout. Test bridge shall be free from defects listed in Table III and other defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory)

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investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Table I and MIL-STD-105. Unit of product shall be one test bridge with conductivity cell.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Sampling for tests shall be in accordance with Table II and MIL-STD-105. Unit of product shall be one test bridge with conductivity cell, and acceptance number shall be zero.

Table II Sampling for tests

Characteristic	Requirement	Test procedure
Performance	3.3	4.4.1
Accuracy	3.4	4.4.1

4.3 Examination. The test bridge shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in Table III.

Table III Classification of defects

Category	Defect
Major	
101	Component part missing.
102	Service data missing.
103	Service data not complete, correct or legible.
104	Marking on dials not complete, correct or legible.
105	Dials not clear.
106	Dials not free of cracks.
107	Case not free of sharp edges.
108	Electrodes not free of cracks.

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Classification of defects (cont'd)

Category Minor	Defect
201	Case surface not smooth, not free of gouges, cracks or scratches.
202	Dials not free of scratches.
203	Service data not free of tears and wrinkles.
204	Identification marking not present, complete, correct, legible or permanent.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* **4.4.1 Performance and accuracy. Test bridge with conductivity cell shall be checked against an accurate instrument whose calibration is traceable to National Bureau of Standards standards. Readings shall be checked and recorded at 500, 1500 and 7000 micromhos at 64° and 190°F. in accordance with procedures in ASTM Designation D1125.**

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Level C. The subject commodity shall be packaged and packed in substantial commercial container (export when applicable) of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

* 5.2 Marking.

5.2.1 Interior packages. Each package shall be marked with National Stock Number, item description, quantity, unit of issue, date packed, contract or purchase order number, and name of contractor.

5.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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* 6.2 This specification covers the following item shown in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-678-5582	TEST, BRIDGE

* 6.3 This specification does not include all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet military requirements.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6630-0219

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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